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AWR 201

Process Work 2

First Topic: Treatment of alcoholism both medically and psychologically.

1. I'm very interested in the treatment and cause of addiction. My major is in biochemistry, I'm looking towards job's/research into addiction.
2. Alcoholism also referred to as alcohol use disorder (AUD) affects roughly 29 million Americans, 209 million people globally. In the U.S. alone alcohol related death are around 178,000 per year. Even with these staggering numbers only 7% of people with AUD are given any treatment in a given year. Additionally, is the high relapse rate known around 40%-60% in the first year (I need to do more research to make sure). Effective medications like naltrexone and acamprosate exist but are dramatically under prescribed, while peer-support models like AA are the cultural default despite mixed evidence on their effectiveness. The gap between what science supports and what patients actually receive is debated in addiction medicine.
3.
 - Does combining pharmacological treatment with cognitive-behavioral therapy produce better long-term abstinence outcomes than either approach alone?
 - Why are FDA-approved medications for AUD (naltrexone, acamprosate, disulfiram) so underprescribed by primary care physicians?
 - How do the neurological changes caused by chronic alcohol use shape relapse risk even after extended abstinence, in order to understand why AUD behaves as a chronic relapsing condition?
 - Do harm-reduction approaches (controlled drinking) produce better outcomes for certain patient subgroups than abstinence-only approaches?
4. Most directly, the 29 million U.S. adults with AUD and their families. But the impact is unevenly distributed. Men die of alcohol related causes at roughly 2–3 times the rate of women, though the gap is narrowing as women's drinking rises. Native American and Alaska Native populations face the highest alcohol related mortality rates of any U.S. demographic group. Adults aged 45–74 have the heaviest mortality rate. Beyond the patients themselves, the topic effects healthcare systems (alcohol is a leading cause of preventable ER visits and liver transplants), employers, the criminal justice system, and children of parents with AUD who face elevated risk of developing AUD themselves.
5. There are a lot of different viewpoints on what addiction is and how to treat it here are the most popular:
 - Disease model vs. behavioral/choice model: Is addiction a chronic brain disease or a learned behavior responsive to incentives and environment?
 - Abstinence-only vs. harm reduction: AA and most U.S. treatment centers require total abstinence, while European models accept moderation as an outcome (methadone for opioid users).

- Medication vs. psychosocial treatment: Should pharmacology be first-line, or are behavioral interventions sufficient for most patients?
 - 12-step vs. evidence-based approaches: AA is the most accessible support system but has historically resisted clinical evaluation. Researchers argue its dominance has stopped better-supported methods chance.
6. General Topic: Addiction Treatment
 Specified Research Topic: Medical and psychological treatment of alcohol use disorder
 Keywords: alcoholism, alcohol use disorder; AUD; pharmacotherapy; cognitive-behavioral therapy; relapse prevention; Alcoholics Anonymous, AA.
7. Specified Research Topic: Medical and psychological treatment of alcohol use disorder
 Primary Source: Treatment for Alcohol Problems: Finding and Getting Help — National Institute on Alcohol Abuse and Alcoholism (NIAAA), NIH. <https://www.niaaa.nih.gov/publications/brochures-and-fact-sheets/treatment-alcohol-problems-finding-and-getting-help>
 This is an NIAAA publication a federal government primary source that lays out the current evidence-based treatment options for AUD. It covers both medications and behavioral approaches to treat AUD.

Second Topic: F1, sustainable fuels, and the future of the internal combustion engine?

1. I love watching F1, following all the races, all the drama. Recently there was a rule change to incorporate more electricity into the cars and use of 100% sustainable fuels. I'm wondering does F1 impact go outside of racing and what does that mean for other car manufactures.
2. F1's 2026 regulations mandate 100% sustainable fuels and a roughly 50/50 split between internal combustion and electric power, which Formula 1 leadership has explicitly done to demonstration that ICE vehicles have a future. This is against the EU's planned 2035 ban on new ICE passenger car sales, California's 2035 zero-emission mandate, and similar policies in the UK, Norway, and parts of Asia.
3.
 - Does F1's sustainable fuel program produce fuels that are technically and economically viable for mass-market passenger car use?
 - Why have major automakers committed to F1's 2026 sustainable-fuel hybrid platform at the same moment most are also committing to all-EV roadmaps?
 - Has F1's historical role as a technology transfer pipeline to consumer cars (disc brakes, hybrid systems, aerodynamics) genuinely continued into the sustainable-fuel era, or has the transfer become primarily symbolic?
 - How does the lifecycle carbon footprint of F1-developed sustainable fuels compare to that of battery electric vehicles?
4. Most Directly, everyone who drives a car, who relays on cars for food, goods, transportation, enjoyment. Basically, everyone in the world. roughly 1.4 billion passenger vehicles currently on the road globally, virtually all of which use internal combustion. If sustainable fuels prove viable, those vehicles could

potentially be decarbonized without replacement; if not, they'll need to be phased out and replaced. Beyond existing car owners, the impact spreads to automakers, oil and energy companies, everyone relies on cars and engines in our modern world, everyone would be affected. Cars are just a step towards a sustainable economy.

5.

- Sustainable fuels as a legitimate parallel path: The e-fuel industry argues e-fuels can decarbonize the existing fleet faster than replacing it, since you can't replace 1.4 billion cars in a decade.
- The motorsport-relevance perspective: Some argue F1's fuel work is genuinely transferable to road use, while others say the 2026 regulations are about preserving the sport's identity, not advancing road technology.
- Sustainable fuels as industry delay: Environmental groups argue synthetic fuels are too expensive, too energy-inefficient, and primarily serve to extend the relevance of ICE manufacturers and oil companies past the point where electrification is ready.
- The thermodynamic critique: Engineers argue that using renewable electricity to make synthetic fuels and then burning them at ~30% efficiency is dramatically worse than putting that same electricity directly into a battery at ~80% efficiency.
- The grid counter: The argument that battery manufacturing's mineral (lithium, cobalt, nickel), grid carbon intensity in many regions, and infrastructure constraints make EVs less clean than numbers suggest.

6. General Topic: Cars Specified Research

Topic: F1's sustainable fuels program and the future of the internal combustion engine

Keywords: Formula 1; sustainable fuels; e-fuels; synthetic fuels; 2026 regulations; internal combustion engine; ICE ban; electrification.

7. Specified Research Topic: F1's sustainable fuels program and the future of the internal combustion engine

Primary Source: "2026 F1 power unit regulations explained: More electrical power, more overtaking aids and more sustainable" — Formula 1 <https://www.formula1.com/en/latest/article/explained-2026-power-unit-regulations-fia.68izKQ2tn1voQPWvgLVMXN>

Directly published by F1, the article specifically explains why F1 is moving to 100% sustainable fuel and frames the decision as deliberately road relevant.